
SAT MATH

Solved Exercises

September 2026

20 fully solved and verified exercises
Algebra · Advanced Math · Problem Solving & Data Analysis · Geometry &
Trigonometry

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Question 1

Question 1

In the xy -plane, consider the parabola $y = x^2 + px + 6$, where p is a constant. Using Desmos with adjustable sliders in the form $y = x^2 - Sx + G$, it was found that the curve matches the original parabola exactly when $S = 7$ and $G = 6$. What is the value of p ?

Solution & Verification

If the two parabolas coincide, their coefficients must be equal:

$$x^2 + px + 6 = x^2 - Sx + G$$

So $G = 6$ (confirming the constant term) and $p = -S$.

With $S = 7$:

$$p = -7$$

Verification: The equation becomes $y = x^2 - 7x + 6 = (x - 1)(x - 6)$, with roots $x = 1$ and $x = 6$. Indeed, the sum of the roots $= 7 = -p$ and the product of the roots $= 6$, matching the constant term.

✓

Answer: $p = -7$

Question 2

Question 2

The functions f and g are defined by

$$f(x) = x^2 + px + 42 \quad \text{and} \quad g(x) = x^2 + 10x + 18,$$

where p is a constant. The solutions to $f(x) = 0$ are r and s , and the solutions to $g(x) = 0$ are $r + 2$ and $s + 2$. What is the value of p ?

Solution & Verification

By Vieta's formulas for f : $r + s = -p$ and $rs = 42$.

By Vieta's formulas for g :

$$(r + 2) + (s + 2) = -10 \Rightarrow r + s = -14$$

$$(r + 2)(s + 2) = 18 \Rightarrow rs + 2(r + s) + 4 = 18$$

Substituting $r + s = -14$:

$$rs - 28 + 4 = 18 \Rightarrow rs = 42$$

This matches the constant term of f (consistency check ✓).

Therefore:

$$p = -(r + s) = -(-14) = 14$$

Verification (SymPy): Solving the full system of 4 equations simultaneously gives $p = 14$ (with $r, s = -7 \mp \sqrt{7}$), confirming the result.

Answer: $p = 14$

Question 3

Question 3

$$f(x) = 3x^2 + 48x + 193$$

The function g is defined by $g(x) = f(x + 7)$. What is the minimum value of $g(x)$?

- A) -15
- B) -8
- C) 1
- D) 8

Solution & Verification

$g(x) = f(x + 7)$ is simply a horizontal shift of f by 7 units. A horizontal shift **does not change** the minimum value of the function — it only moves the location where it occurs.

So it suffices to find the minimum value of f . Vertex at:

$$x = -\frac{b}{2a} = -\frac{48}{2 \cdot 3} = -8$$

$$f(-8) = 3(64) + 48(-8) + 193 = 192 - 384 + 193 = 1$$

Verification: Fully expanding $g(x) = f(x + 7) = 3(x + 7)^2 + 48(x + 7) + 193$ and finding its vertex again gives a minimum of 1 (at $x = -15$).

Answer: C) 1

Question 4

Question 4

A sample of a certain isotope takes 29 years to decay to half its original mass. The function $s(t) = 128(0.5)^{t/29}$ gives the approximate mass of this isotope, in grams, that remains t years after a 128-gram sample starts to decay. Which statement is the best interpretation of $s(58) = 32$ in this context?

- A) Approximately 32 grams of the sample remains 58 years after the sample starts to decay.
- B) The mass of the sample has decreased by approximately 32 grams 58 years after the sample starts to decay.
- C) The mass of the sample has decreased by approximately 58 grams 32 years after the sample starts to decay.
- D) Approximately 58 grams of the sample remains 32 years after the sample starts to decay.

Solution & Verification

In the relation $s(t) = 32$, $t = 58$ is the *input* (time in years) and 32 is the *output* (mass in grams that **remains**, not that was lost).

Verification by substitution:

$$s(58) = 128(0.5)^{58/29} = 128(0.5)^2 = 128 \times 0.25 = 32 \checkmark$$

So the correct interpretation is: after 58 years, approximately 32 grams remain (not that 32 grams were lost).

Answer: A

Question 5

Question 5

$$x(13x + 27) + c = 5$$

In the given equation, c is a constant. A solution to the equation is $\frac{-27 + \sqrt{261}}{26}$. What is the value of c ?

- A) 14
- B) 9
- C) 5
- D) 4

Solution & Verification

Expanding: $13x^2 + 27x + (c - 5) = 0$.

The quadratic formula gives:

$$x = \frac{-27 \pm \sqrt{27^2 - 4(13)(c - 5)}}{26} = \frac{-27 \pm \sqrt{729 - 52(c - 5)}}{26}$$

Comparing with the given solution $\frac{-27 + \sqrt{261}}{26}$, we need:

$$729 - 52(c - 5) = 261 \Rightarrow 52(c - 5) = 468 \Rightarrow c - 5 = 9 \Rightarrow c = 14$$

Verification: With $c = 14$, the equation becomes $13x^2 + 27x + 9 = 0$, with discriminant $729 - 468 = 261$, which exactly matches the given root. ✓

Answer: A) 14

Question 6

Question 6

The function f is defined by $f(x) = \frac{|x|}{a} - 14$, where $a < 0$. What is the product of $f(15a)$ and $f(8a)$?

Solution & Verification

Since $a < 0$, both $15a$ and $8a$ are also negative, so $|15a| = -15a$ and $|8a| = -8a$.

$$f(15a) = \frac{-15a}{a} - 14 = -15 - 14 = -29$$

$$f(8a) = \frac{-8a}{a} - 14 = -8 - 14 = -22$$

Product:

$$f(15a) \cdot f(8a) = (-29)(-22) = 638$$

Verification: The result is independent of the specific value of a (as long as $a < 0$), which was confirmed symbolically with SymPy.

Answer: 638

Question 7

Question 7

On a plot of land, 52.0% of the square footage is farmland and the remaining square footage is pasture. There are buildings on exactly 21.5% of the square footage of the farmland, and there are buildings on exactly 14.0% of the square footage of the pasture. If there are buildings on exactly $p\%$ of the square footage of the plot of land, what is the value of p ?

Solution & Verification

Let the total area of the plot of land be 100 (units, without loss of generality).

- Farmland: 52.0, with buildings on 21.5% of it $\Rightarrow 0.52 \times 21.5 = 11.18$
- Pasture: $100 - 52 = 48.0$, with buildings on 14.0% of it $\Rightarrow 0.48 \times 14.0 = 6.72$

Total percentage with buildings:

$$p = 11.18 + 6.72 = 17.9$$

Verification: $11.18 + 6.72 = 17.90$ exactly, so the result does not need rounding.

Answer: $p = 17.9$

Question 8

Question 8

$$y < 24 - 8x$$

$$\frac{y}{8} > 9$$

Which inequality represents the x values for all solutions (x, y) that satisfy the given system of inequalities in the xy -plane?

- A) $x > 3$
- B) $x < 3$
- C) $x > -6$
- D) $x < -6$

Solution & Verification

From the second inequality: $\frac{y}{8} > 9 \Rightarrow y > 72$.

For a common solution to exist, there must be some y satisfying both $y > 72$ and $y < 24 - 8x$ simultaneously, so:

$$72 < 24 - 8x$$

$$8x < 24 - 72 = -48$$

$$x < -6$$

Verification: For $x = -7$ (satisfying $x < -6$): $24 - 8(-7) = 80 > 72$, so there exists a y (e.g. $y = 75$) satisfying both inequalities ✓. For $x = -5$ (outside the solution): $24 - 8(-5) = 64 < 72$, which is impossible.

Answer: D) $x < -6$

Question 9

Question 9

$$F(x) = \frac{9}{5}(x - 273.15) + 32$$

The function F gives the temperature, in degrees Fahrenheit, that corresponds to x kelvins. If a temperature is increased by 11.70 kelvins, by how much did the temperature increase in degrees Fahrenheit?

Solution & Verification

F is a linear function of x with slope $\frac{9}{5}$. The increase in Fahrenheit equals the slope times the increase in Kelvin (the constant $-273.15 \cdot \frac{9}{5} + 32$ does not affect the *change*):

$$\Delta F = \frac{9}{5} \times \Delta x = \frac{9}{5} \times 11.70 = 21.06$$

Verification: $F(x_0 + 11.70) - F(x_0) = \frac{9}{5}(x_0 + 11.70 - 273.15) + 32 - [\frac{9}{5}(x_0 - 273.15) + 32] = \frac{9}{5}(11.70) = 21.06$, independent of x_0 . ✓

Answer: 21.06 °F

Question 10

Question 10

A space probe uses a square-shaped solar sail to move through space. The side length of the solar sail is $8.89w$ meters, where w is the width, in meters, of the space probe. Which equation gives the area A , in square meters, of the solar sail?

- A) $A = (w + 8.89)(w + 8.89)$
- B) $A = (4)(8.89w)$
- C) $A = (8.89w)(8.89w)$
- D) $A = \left(\frac{8.89}{w}\right)\left(\frac{8.89}{w}\right)$

Solution & Verification

The area of a square with side ℓ is $A = \ell^2 = \ell \cdot \ell$.

Here $\ell = 8.89w$, so:

$$A = (8.89w)(8.89w)$$

Choice B incorrectly corresponds to the perimeter ($4 \times \text{side}$) rather than the area. Choices A and D distort the expression for the side entirely.

Answer: C) $A = (8.89w)(8.89w)$

Question 11

Question 11

When several resistors are connected in series in a circuit, the total resistance of the circuit is the sum of the resistances of these resistors. A certain circuit consists of 7 resistors connected in series, where the resistance of each resistor is positive. This circuit has a total resistance of 80 ohms. The total resistance of 3 resistors in this circuit is 60 ohms. Which inequality best represents all possible values of the resistance x , in ohms, of one of the other 4 resistors in this circuit?

- A) $20 < x < 80$
- B) $20 < x < 60$
- C) $0 < x < 20$
- D) $0 < x < 5$

Solution & Verification

The remaining 4 resistors have a combined total of:

$$80 - 60 = 20 \text{ ohms}$$

Let x be the resistance of one of these 4, and the other 3 (of these 4) have a combined total of $20 - x$. Since **every** resistance must be **positive**:

$$x > 0 \quad \text{and} \quad 20 - x > 0 \Rightarrow x < 20$$

Therefore:

$$0 < x < 20$$

Verification: If $x = 20$ or greater, the remaining 3 resistors would need a combined total ≤ 0 , which is impossible since all are positive.

Answer: C) $0 < x < 20$

Question 12

Question 12

A moving truck rental company charges its customers \$165 for the first hour of renting a truck and \$95 for each additional hour, plus the cost of gas used. Which equation represents this situation, where y is the total cost, in dollars, of renting a truck from this company for x hours, w is the cost, in dollars, of gas used, and $x \geq 1$?

- A) $y = 95(x - 1) + w$
- B) $y = 165 + 95 + w$
- C) $y = 165 + 95x + w$
- D) $y = 165 + 95(x - 1) + w$

Solution & Verification

The first hour costs a flat \$165. The **additional** hours are $x - 1$ (since the first hour is already counted), each costing \$95:

$$y = 165 + 95(x - 1) + w$$

Verification: For $x = 1$ (only the first hour): $y = 165 + 95(0) + w = 165 + w$ ✓. For $x = 3$: $y = 165 + 95(2) + w = 355 + w$, i.e. \$165 for the first hour plus $2 \times \$95$ for the other two. Choice C incorrectly charges \$95 for the first hour as well.

Answer: D) $y = 165 + 95(x - 1) + w$

Question 13

Question 13

For a party, 41 hamburger buns are needed. Hamburger buns are sold in packages of 8. What is the minimum number of packages that should be bought for the party?

Solution & Verification

$$\frac{41}{8} = 5.125$$

Since buns are sold only in full packages, we need the **next integer up** (round up):

$$\lceil 5.125 \rceil = 6$$

Verification: 5 packages give only $5 \times 8 = 40 < 41$ (not enough). 6 packages give $6 \times 8 = 48 \geq 41$ ✓.

Answer: 6 packages

Question 14

Question 14

The population density of Town A is 120 people per square mile, and the population density of Town B is 90 people per square mile. Town A has a population of 20,640 people, and Town B has a population of 3,870 people. The area, in square miles, of Town A is k times the area, in square miles, of Town B. What is the value of k ?

Solution & Verification

$$\text{Area} = \frac{\text{Population}}{\text{Density}}.$$

$$\text{Area}_A = \frac{20640}{120} = 172 \text{ sq mi}$$

$$\text{Area}_B = \frac{3870}{90} = 43 \text{ sq mi}$$

$$k = \frac{\text{Area}_A}{\text{Area}_B} = \frac{172}{43} = 4$$

Verification: $43 \times 4 = 172 \checkmark$.

Answer: $k = 4$

Question 15

Question 15

The height of an equilateral triangle is $52\sqrt{3}$ feet. What is the perimeter, in feet, of the equilateral triangle?

Solution & Verification

For an equilateral triangle with side a , the height is $h = \frac{\sqrt{3}}{2}a$.

$$\frac{\sqrt{3}}{2}a = 52\sqrt{3} \Rightarrow a = 52\sqrt{3} \cdot \frac{2}{\sqrt{3}} = 104$$

Perimeter:

$$P = 3a = 3 \times 104 = 312$$

Verification: With side 104, height $= \frac{\sqrt{3}}{2}(104) = 52\sqrt{3}$ ✓, matching the given value.

Answer: 312 feet

Question 16

Question 16

$$y = (x + 18)(x^2 - 25)$$

In the xy -plane, the graph of the given equation contains the point (s, t) . If $-5 \leq s \leq 5$, which of the following is NOT a possible value of t ?

- A) 18
- B) 5
- C) 0
- D) -5

Solution & Verification

For $-5 \leq x \leq 5$ we have $0 \leq x^2 \leq 25$, so the factor $(x^2 - 25)$ is **always** in the interval $[-25, 0]$ (non-positive).

Also, the factor $(x + 18)$ for $x \in [-5, 5]$ takes values in $[13, 23]$ (always positive).

So the product $t = (x + 18)(x^2 - 25)$ is **always** ≤ 0 on this interval, with $t = 0$ only at the endpoints $x = \pm 5$.

Computing the minimum (derivative = 0 at $x \approx 0.658$): $t_{\min} \approx -458.4$. So the actual range of values is $t \in [-458.4, 0]$.

Verification note: Mathematically, both 18 and 5 fall **outside** the achievable range $[-458.4, 0]$ (since they are positive numbers), so strictly speaking both choices A and B are not achievable. As the intended single-choice answer, A is selected (the more extreme value), but this is flagged explicitly so it can be checked against the original source — a possible inconsistency/typo in the problem statement (e.g. a different bound on s , or a sign on the 18).

Answer: A) 18 (see verification note)

Question 17

Question 17

$$x^2 - 14x + y^2 + 22y = 26$$

In the xy -plane, the graph of the given equation is a circle. What is the length of the circle's diameter?

- A) 13
- B) 28
- C) 52
- D) 196

Solution & Verification

Completing the square:

$$(x^2 - 14x + 49) + (y^2 + 22y + 121) = 26 + 49 + 121$$

$$(x - 7)^2 + (y + 11)^2 = 196$$

So $r^2 = 196 \Rightarrow r = 14$, and the diameter:

$$d = 2r = 28$$

Verification: $26 + 49 + 121 = 196 = 14^2 \checkmark$.

Answer: B) 28

Question 18

Question 18

The function $f(n) = 7(20.52)^{n/4}$ gives each term of a sequence as a function of the term's position, n , in the sequence, where n is a whole number. The value of the term in position 14 is $p\%$ more than the value of the term in position 10. What is the value of p ?

- A) 2,052
- B) 1,952
- C) 52

Solution & Verification

$$\frac{f(14)}{f(10)} = \frac{7(20.52)^{14/4}}{7(20.52)^{10/4}} = (20.52)^{(14-10)/4} = (20.52)^1 = 20.52$$

The percent increase:

$$p = (20.52 - 1) \times 100 = 19.52 \times 100 = 1952$$

Verification: The ratio $f(14)/f(10) = 20.52$ means the term at position 14 is 20.52 times the term at position 10, i.e. 1952% *more* (not 2052%, which would simply be 20.52 times 100 without subtracting the original value).

Answer: B) 1,952

Question 19

Question 19

In isosceles triangle ABC , sides AB and AC are congruent. Point D divides side BC such that the length of segment BD is three fourths of the length of segment BC . Point E lies on side AB and point F lies on side AC such that when segments DE and DF are drawn, angle BED is congruent to angle CFD . If the length of segment BE is 12, what is the length of segment CF ?

Solution & Verification

Since the triangle is isosceles with $AB = AC$, the base angles are equal: $\angle B = \angle C$. We are also given $\angle BED \cong \angle CFD$. So, in triangles BED and CFD , two angles are respectively equal ($\angle B = \angle C$ and $\angle BED = \angle CFD$), so (AA criterion) the triangles are **similar**:

$$\triangle BED \sim \triangle CFD$$

with vertex correspondence $B \leftrightarrow C, E \leftrightarrow F, D \leftrightarrow D$.

So the corresponding sides are proportional:

$$\frac{BE}{CF} = \frac{BD}{CD}$$

Given $BD = \frac{3}{4}BC$, so $CD = BC - BD = \frac{1}{4}BC$, hence:

$$\frac{BD}{CD} = \frac{3/4}{1/4} = 3$$

Therefore:

$$\frac{12}{CF} = 3 \Rightarrow CF = \frac{12}{3} = 4$$

Verification: With $BC = 4$ (units), $BD = 3$, $CD = 1$: the similarity ratio is $3 : 1$, so $CF = BE/3 = 12/3 = 4 \checkmark$.

Answer: $CF = 4$

Question 20

Question 20

A giant panda is predicted to have a mass between $\frac{1}{950}$ and $\frac{1}{850}$ times its mother's mass when it is born. If its mother is predicted to have a mass of 112,100 grams when the giant panda is born, which inequality best represents the prediction of the giant panda's mass p , in grams, when it is born?

- A) $11.8 \leq p < 13.189$
- B) $118 \leq p < 131.89$
- C) $1,180 \leq p < 1,318.9$
- D) $11,800 \leq p < 13,189$

Solution & Verification

Lower bound:

$$\frac{112100}{950} = 118$$

Upper bound:

$$\frac{112100}{850} \approx 131.882$$

Therefore:

$$118 \leq p < 131.89$$

Verification: $950 \times 118 = 112100$ ✓ (exact). $850 \times 131.89 \approx 112101.5 \approx 112100$ ✓.

Answer: B) $118 \leq p < 131.89$

Question 21

Question 21

The figure shown is a right rectangular pyramid, where $\ell = 18$ units, $w = 9$ units, and $h = 12$ units. What is the surface area, in square units, of the pyramid?

Solution & Verification

The pyramid has a rectangular base $\ell \times w$ and four triangular lateral faces (two pairs, since opposite faces are congruent).

Base area:

$$\ell \cdot w = 18 \times 9 = 162$$

Slant height of the two faces with base $\ell = 18$ (these faces rise over the w -direction, so their slant height uses h and $w/2$):

$$s_w = \sqrt{h^2 + \left(\frac{w}{2}\right)^2} = \sqrt{12^2 + 4.5^2} = \sqrt{144 + 20.25} = \sqrt{164.25} = \frac{3\sqrt{73}}{2}$$

Area of these two faces combined: $\ell \cdot s_w = 18 \cdot \frac{3\sqrt{73}}{2} = 27\sqrt{73}$.

Slant height of the two faces with base $w = 9$ (these rise over the ℓ -direction):

$$s_\ell = \sqrt{h^2 + \left(\frac{\ell}{2}\right)^2} = \sqrt{12^2 + 9^2} = \sqrt{144 + 81} = \sqrt{225} = 15$$

Area of these two faces combined: $w \cdot s_\ell = 9 \times 15 = 135$.

Total surface area:

$$162 + 27\sqrt{73} + 135 = 297 + 27\sqrt{73} \approx 527.69$$

Verification: Computed independently with SymPy: $297 + 27\sqrt{73} \approx 527.688$, matching the manual calculation. ✓

Answer: $297 + 27\sqrt{73} \approx 527.69$ square units

Question 22

Question 22

In a certain ant colony, 10% of the ants are queen ants. Of the remaining ants, 60% are female worker ants, and the remaining ants are male ants. If there are 2,196 male ants in the colony, what is the total number of female worker ants in this colony?

Solution & Verification

Let T be the total number of ants. Queens = $0.10T$, so the remaining ants = $0.90T$.
Of these, 60% are female workers and 40% are male:

$$\text{males} = 0.40 \times 0.90T = 0.36T$$

Setting this equal to the given number of males:

$$0.36T = 2196 \Rightarrow T = 6100$$

Female workers:

$$0.60 \times 0.90 \times 6100 = 0.54 \times 6100 = 3294$$

Verification: Queens = $0.10(6100) = 610$; males = 2196; females = 3294. Sum: $610 + 2196 + 3294 = 6100 = T$ ✓.

Answer: 3294 female worker ants

Question 23

Question 23

In triangle QRS , angle R is a right angle, $RS = 49$, and $QR < RS$. Which expression represents the length of segment QS ?

- A) $49 \cos Q$
- B) $49 \sin Q$
- C) $\frac{49}{\cos Q}$
- D) $\frac{49}{\sin Q}$

Solution & Verification

Since angle R is the right angle, QS is the hypotenuse. Relative to angle Q : side RS (length 49) is **opposite** angle Q , and side QR is **adjacent** to it.

Using the sine ratio:

$$\sin Q = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{RS}{QS} = \frac{49}{QS}$$

Solving for QS :

$$QS = \frac{49}{\sin Q}$$

Verification: Since $QR < RS$, angle Q is the larger acute angle (opposite the longer leg RS), consistent with the figure. The relation $QS \sin Q = RS = 49$ always holds in a right triangle regardless of the specific angle measure. ✓

Answer: D) $\frac{49}{\sin Q}$

Question 24

Question 24

$$-4x(6x + p) = 54$$

In the given equation, p is an integer constant. The equation has no real solution. What is the least possible value of p ?

Solution & Verification

Expanding:

$$-24x^2 - 4px = 54 \Rightarrow 24x^2 + 4px + 54 = 0$$

For a quadratic $ax^2 + bx + c = 0$ to have **no real solution**, its discriminant must be negative:

$$b^2 - 4ac < 0$$

$$(4p)^2 - 4(24)(54) < 0$$

$$16p^2 - 5184 < 0$$

$$p^2 < 324$$

$$-18 < p < 18$$

Since p must be an **integer**, the least possible value is:

$$p = -17$$

Verification: With $p = -17$: discriminant $= 16(289) - 5184 = 4624 - 5184 = -560 < 0$ ✓ (no real solution). With $p = -18$ (excluded boundary): discriminant $= 16(324) - 5184 = 0$, giving exactly one real solution, not zero solutions — confirming -18 itself does not work and -17 is indeed the least valid integer.

Answer: $p = -17$

Question 25

Question 25

$$f(x) = -\frac{1}{8}(x+18)(x^2-16)$$

For the function f , where $-4 \leq x \leq 4$, which of the following cannot be a value of $f(x)$?

- A) 4
- B) 84
- C) 0
- D) -4

Solution & Verification

Expanding: $f(x) = -\frac{1}{8}x^3 - \frac{9}{4}x^2 + 2x + 36$.

Evaluating at the endpoints: $f(-4) = 0$ and $f(4) = 0$.

Taking the derivative and finding critical points in $[-4, 4]$: there is one critical point at $x \approx 0.434$, where $f(x) \approx 36.43$ (a local maximum). Checking the sign of $f'(x)$ shows f is increasing on $[-4, 0.434]$ and decreasing on $[0.434, 4]$.

So on $[-4, 4]$, $f(x)$ ranges over $[0, 36.43]$ approximately — it never goes negative and never exceeds about 36.43.

- 4 is inside $[0, 36.43] \Rightarrow$ possible.
- 0 occurs at $x = \pm 4 \Rightarrow$ possible.
- -4 is **below** the range $[0, 36.43] \Rightarrow$ **not possible**.
- 84 is also **above** the range $[0, 36.43] \Rightarrow$ technically also not possible.

Verification note: Strictly speaking, both 84 and -4 fall outside the achievable range $[0, 36.43]$, so mathematically neither could occur. This matches the answer marked in your source (D), which is also the more clearly-intended distractor (a value below the minimum). Flagging this the same way as the earlier $(x+18)(x^2-25)$ question, in case the original answer choices had a typo.

Answer: D) -4 (see verification note)

Question 26

Question 26

Jack is planning to build a raised garden bed in the shape of a right triangle. The length of the longer leg is 70 units greater than twice the length of the shorter leg. The function a gives the area, in square units, of the garden bed when the longer leg has a length of x units, where $x > 70$. Which equation defines a ?

- A) $a(x) = \frac{1}{2}x^2 - 35x$
- B) $a(x) = \frac{1}{4}x^2 - \frac{35}{2}x$
- C) $a(x) = x^2 + 35x$
- D) $a(x) = \frac{1}{4}x^2 + \frac{35}{2}x$

Solution & Verification

Let the shorter leg have length s . The longer leg is $x = 2s + 70$, so:

$$s = \frac{x - 70}{2}$$

Area of a right triangle = $\frac{1}{2}(\text{leg}_1)(\text{leg}_2)$:

$$a(x) = \frac{1}{2} \cdot x \cdot \frac{x - 70}{2} = \frac{x(x - 70)}{4} = \frac{x^2 - 70x}{4} = \frac{1}{4}x^2 - \frac{70}{4}x = \frac{1}{4}x^2 - \frac{35}{2}x$$

Verification: For $x = 90$ (so $s = (90 - 70)/2 = 10$): area should be $\frac{1}{2}(90)(10) = 450$. Using the formula: $\frac{1}{4}(8100) - \frac{35}{2}(90) = 2025 - 1575 = 450 \checkmark$.

$$\text{Answer: B) } a(x) = \frac{1}{4}x^2 - \frac{35}{2}x$$

Question 27

Question 27

Right circular cylinders A and B are similar. The height of cylinder A is 10 units, and the diameter of its base is equal to its height. The ratio of the diameter of the base of cylinder A to the diameter of the base of cylinder B is $3 : 7$. What is the volume, in cubic units, of cylinder B ?

- A) 250π
- B) $\frac{1750\pi}{3}$
- C) $\frac{12250\pi}{9}$
- D) $\frac{85750\pi}{27}$

Solution & Verification

For cylinder A : height = 10, and diameter = height = 10, so radius = 5.

Since the cylinders are similar, **all** linear dimensions scale by the same ratio. The diameter ratio $A : B = 3 : 7$ means every length in B is $\frac{7}{3}$ times the corresponding length in A :

$$r_B = 5 \times \frac{7}{3} = \frac{35}{3}, \quad h_B = 10 \times \frac{7}{3} = \frac{70}{3}$$

Volume of a cylinder = $\pi r^2 h$:

$$V_B = \pi \left(\frac{35}{3}\right)^2 \left(\frac{70}{3}\right) = \pi \cdot \frac{1225}{9} \cdot \frac{70}{3} = \frac{85750\pi}{27}$$

Verification: Computed independently with SymPy: $\frac{85750\pi}{27}$, matching exactly. ✓

Answer: D) $\frac{85750\pi}{27}$

Question 28

Question 28

Each face of a fair 16-sided die is labeled with a number from 1 through 16, with a different number appearing on each face. If the die is rolled one time, what is the probability of rolling a 2?

- A) $\frac{15}{16}$
- B) $\frac{14}{16}$
- C) $\frac{2}{16}$
- D) $\frac{1}{16}$

Solution & Verification

There are 16 equally likely, distinct outcomes (one per face), and exactly **one** of them is a 2.

$$P(\text{rolling a 2}) = \frac{\text{favorable outcomes}}{\text{total outcomes}} = \frac{1}{16}$$

Verification: Each face has a unique number from 1 to 16, so the number 2 appears on exactly one face out of sixteen. ✓

Answer: D) $\frac{1}{16}$

Question 29

Question 29

The scatterplot shows the relationship between x and y for a data set. The curve passes through the approximate vertex $(x_0, y_0) = (40, 32)$ and through the point $(30, 24)$; data extends from about $x = 22$ (where $y \approx 5$) to $x = 58$ (where $y \approx 5$), forming a downward-opening, roughly symmetric hump. Which of the following functions could best model the scatterplot?

- A) $y = -\frac{1}{8}(x - 40)^2 + 32$
- B) $y = \frac{1}{8}(x - 40)^2 + 32$
- C) $y = -\frac{1}{8}(x - 32)^2 + 40$
- D) $y = -\frac{1}{4}(x - 40)^2 + 32$

Solution & Verification

The data forms a downward-opening parabola (hump shape), so the leading coefficient must be **negative** — this immediately eliminates choice B (positive coefficient, would open upward).

The peak (vertex) of the data is at approximately $x = 40$, $y = 32$, matching the $(x - 40)^2 + 32$ form in choices A and D, and ruling out choice C (which is centered at $x = 32$, not 40).

To choose between A $(-\frac{1}{8})$ and D $(-\frac{1}{4})$, use a data point away from the vertex, e.g. the left edge of the data near $(22, 5)$:

$$y = a(x - 40)^2 + 32 \Rightarrow 5 = a(22 - 40)^2 + 32 \Rightarrow a = \frac{5 - 32}{324} = \frac{-27}{324} \approx -0.083$$

Comparing to the choices: $-\frac{1}{8} = -0.125$ and $-\frac{1}{4} = -0.25$. The estimated coefficient ≈ -0.083 is much closer to $-\frac{1}{8}$ than to the steeper $-\frac{1}{4}$ (which would drop far too quickly and predict negative y -values well before $x = 22$).

Verification: Checking with $a = -\frac{1}{8}$ at $x = 30$: $y = -\frac{1}{8}(30 - 40)^2 + 32 = -\frac{1}{8}(100) + 32 = -12.5 + 32 = 19.5$, reasonably close to the plotted value near 24, given this is an approximate best-fit model rather than an exact interpolation.

Answer: A) $y = -\frac{1}{8}(x - 40)^2 + 32$

Question 30

Question 30

In 2010, a skating team had a total of 36 skaters, each classified as either advanced or intermediate. From 2010 to 2020, the number of advanced skaters on the team increased by approximately 23%, and the number of intermediate skaters on the team increased by approximately 36%. The total number of skaters on the team increased by approximately 28%. Which equation best represents this situation, where a represents the number of advanced skaters on the team in 2010 and b represents the number of intermediate skaters on the team in 2010?

- A) $1.23a + 1.28b = 36(1.36)$
- B) $1.28a + 0.23b = 36(1.36)$
- C) $1.23a + 1.36b = 36(1.28)$
- D) $1.36a + 1.23b = 36(1.28)$

Solution & Verification

In 2010: $a + b = 36$ (total skaters).

By 2020: the advanced group grew by 23%, becoming $1.23a$; the intermediate group grew by 36%, becoming $1.36b$. Together, these must equal the new total, which is the original 36 increased by 28%:

$$1.23a + 1.36b = 36(1.28)$$

Verification: Each coefficient is correctly paired with its own variable (1.23 with a for advanced, 1.36 with b for intermediate), and the right-hand side correctly reflects the *overall* 28% increase applied to the original total of 36, not to either subgroup individually. ✓

Answer: C) $1.23a + 1.36b = 36(1.28)$

Question 31

Question 31

The graph of $y = g(x)$ is shown, crossing the x -axis once near $x = -2$ and touching it (without crossing) near $x = 1$ and near $x = 3$. If

$$g(x + 10) = \frac{1}{10}(x - a)^2(x - b)^2(x - c),$$

where a , b , and c are constants, which of the following could be the value of b ?

Solution & Verification

Because g touches (but doesn't cross) the axis at $x = 1$ and $x = 3$, those are **double roots** of g ; because g crosses the axis at $x = -2$, that is a **single root**.

The given form $g(x + 10) = \frac{1}{10}(x - a)^2(x - b)^2(x - c)$ has **two squared factors** (double roots, matching $x = 1$ and $x = 3$) and **one single factor** (matching $x = -2$).

If $g(r) = 0$, then $g(x + 10) = 0$ when $x + 10 = r$, i.e. $x = r - 10$. Shifting each root of g by -10 :

$$x = -2 \Rightarrow -2 - 10 = -12 \quad (\text{single root, corresponds to } c)$$

$$x = 1 \Rightarrow 1 - 10 = -9 \quad (\text{double root, corresponds to } a \text{ or } b)$$

$$x = 3 \Rightarrow 3 - 10 = -7 \quad (\text{double root, corresponds to } a \text{ or } b)$$

Since a and b play symmetric (interchangeable) roles in the formula, b could be either of the double-root values:

$$b = -9 \quad \text{or} \quad b = -7$$

Verification: This gives $g(x + 10) = \frac{1}{10}(x + 12)(x + 9)^2(x + 7)^2$, matching the factored form with $c = -12$ and $\{a, b\} = \{-9, -7\}$. ✓

Answer: $b = -9$ or $b = -7$

Question 32

Question 32

The solutions to $x^2 + px + 56 = 0$, where p is a constant, are r and s . The solutions to $x^2 + 12x + 28 = 0$ are $r + 2$ and $s + 2$. What is the value of p ?

Solution & Verification

By Vieta's formulas for the second equation: sum of roots $= -12$, so:

$$(r + 2) + (s + 2) = -12 \Rightarrow r + s = -16$$

Product of roots $= 28$:

$$(r + 2)(s + 2) = 28 \Rightarrow rs + 2(r + s) + 4 = 28$$

$$rs + 2(-16) + 4 = 28 \Rightarrow rs - 28 = 28 \Rightarrow rs = 56$$

This matches the constant term of the first equation (consistency check ✓).

Therefore:

$$p = -(r + s) = -(-16) = 16$$

Verification: With $r + s = -16$ and $rs = 56$, the roots are $r, s = -8 \mp 2\sqrt{2}$ (both real). Substituting back into $x^2 + px + 56 = 0$ with $p = 16$ confirms consistency, matching the value you had already written down.

Answer: $p = 16$